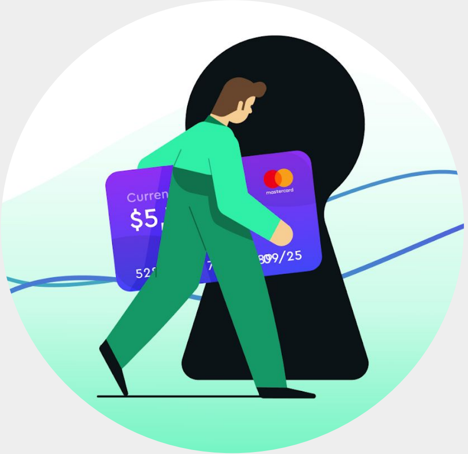

Enhancing Credit Card Fraud Detection through Oversampling:

A Comparative Analysis of SMOTE and VAE

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Introduction

Problem Statement

Detecting credit card fraud accurately while minimizing false positives is challenging due to class imbalance, with far fewer fraudulent transactions than genuine ones.

Proposed Solution

We use VAE and SMOTE to improve existing classification models (Logistic Regression, XGBoost, and MLP) by oversampling the imbalanced training dataset.

Data Preparation

Dataset Overview

Kaggle dataset contains **European credit card transaction data** across two days in September 2013.

248,807 total transactions

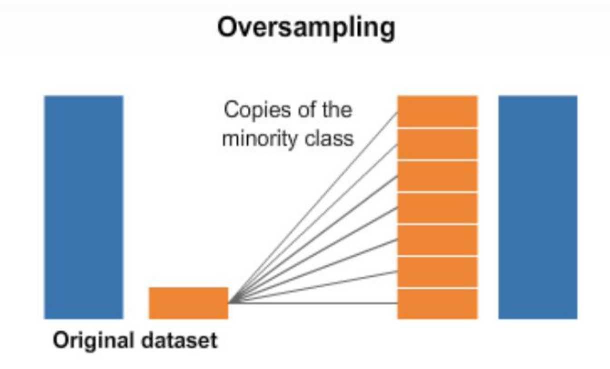
31 Feature Columns:

- 28 numerical variables from PCA principal components
- Time (transaction time)
- Amount (transaction value)
- Class (1: fraudulent, 0: genuine)

Data Processing

1,825 of the transactions (0.64% of dataset) had an **'Amount' of 0**. We chose to **omit these rows**.

Handling Data Imbalance



Class Imbalance

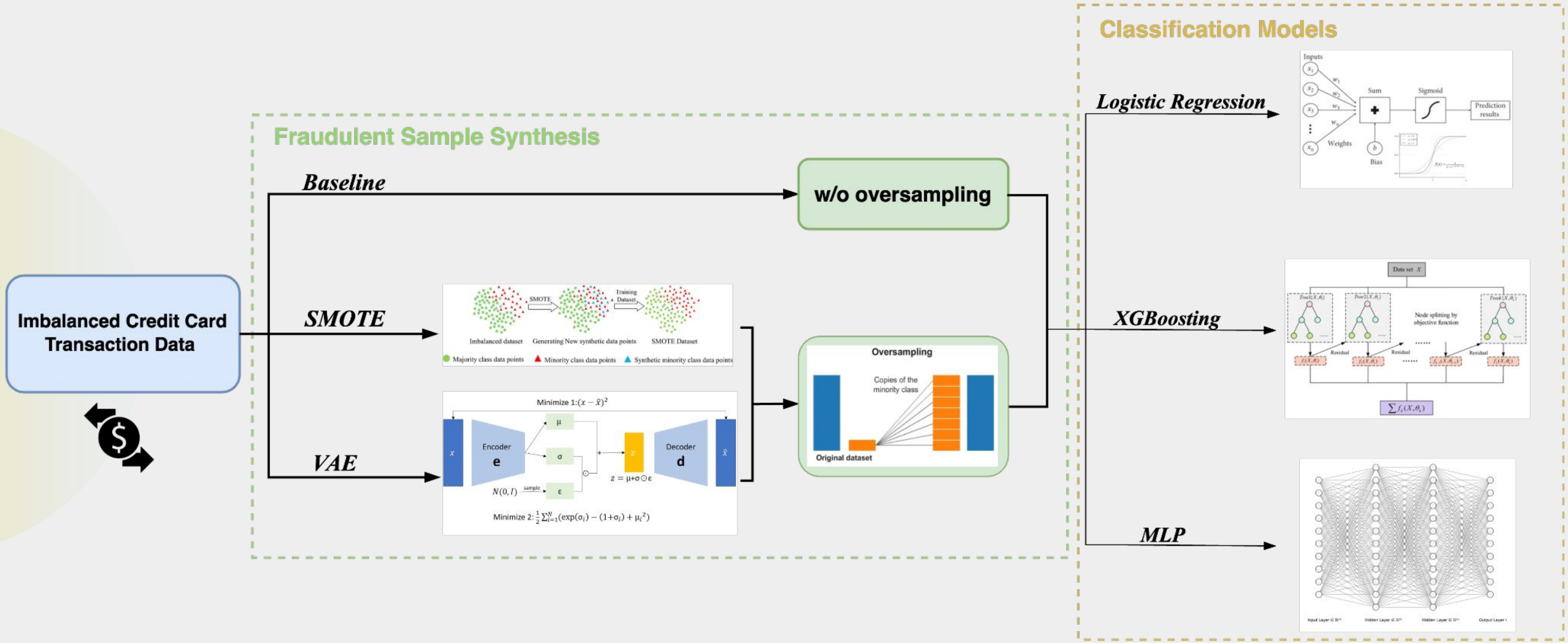
282,517 non-fraudulent vs. **465** fraudulent transactions.

Strategies

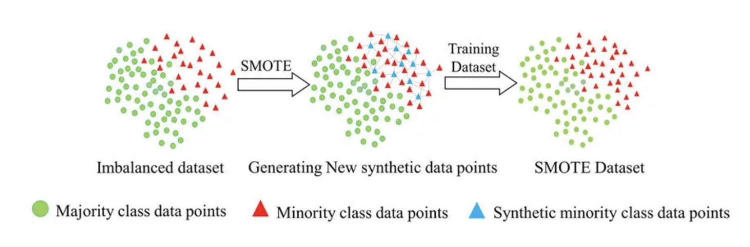
Synthesize "fake" fraudulent samples to address class imbalance.

- **SMOTE:** A commonly used interpolation method that creates synthetic samples for the fraudulent class.
- **VAE:** A deep learning-driven generative model that captures the complex probability distribution of fraudulent transactions.

Fraud Detection System Overview

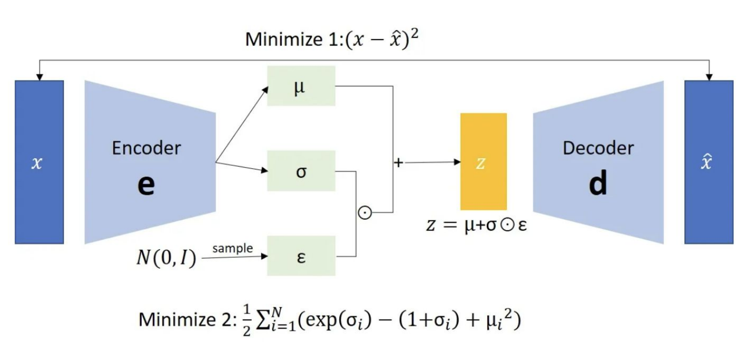


Sampling Component: SMOTE & VAE



Strategy 1: SMOTE

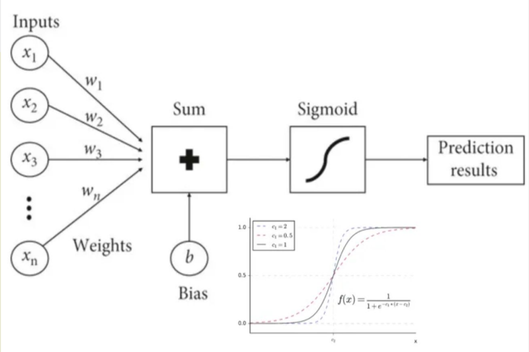
- Identifies nearest neighbors of minority class samples.
- Creates synthetic samples by interpolating between existing minority class instances.
- A simple technique to help balance the dataset by increasing the representation of the fraudulent class.

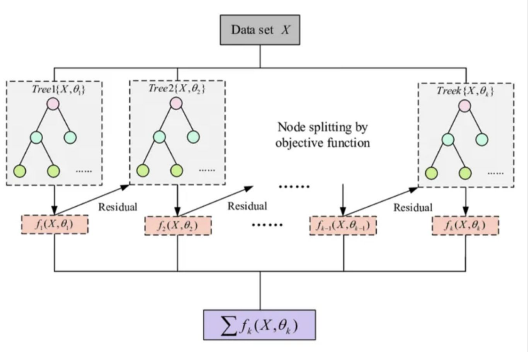


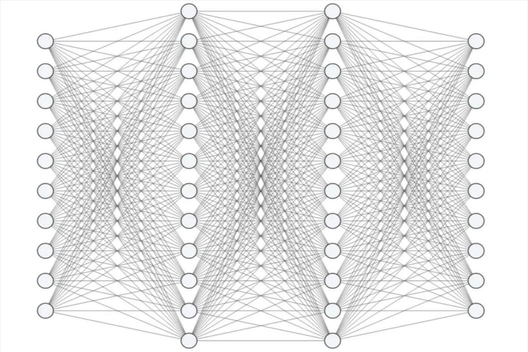
Strategy 2: Variational Autoencoder

- Models the minority class distribution using variational bayes (by training on the minority samples)
- Generates new synthetic minority data points by sampling from the latent space (standard gaussian).
- A deep learning approach to producing more diverse and realistic fraudulent samples.

Classification Component







Logistic Regression

A low-capacity linear classifier that can generalize well and can offer interpretability

XGBoost

An ensemble model class based on gradient boosting, excels with imbalanced data and complex patterns.

Multi-Layer Perceptron

A simple neural network that captures non-linear relationships, ideal for complex datasets.

Model Performance & Evaluation

Want to maximize accuracy, while ensuring a low false-positive rate (incorrect detection of fraud).

Evaluation Metrics: Area Under the Precision-Recall Curve (AUPRC)

	Logistic Regression	XGBoost	MLP
w/o oversampling	0.6543	0.7714	0.6798
w/ oversampling (SMOTE)	0.7452	0.8129	0.7685
w/ oversampling (VAE)	**0.8046**	**0.8932**	**0.8347**

Conclusion

VAE outperformed SMOTE

The Variational Autoencoder (VAE) demonstrated superior performance as an oversampling technique. It better captures fraudulent data distribution, demonstrating the neural network's ability to generate more realistic and diverse samples.

XGBoost achieved the best results

Among the models tested, XGBoost achieved the highest AUPRC, effectively handling complex patterns and imbalanced data with VAE. MLP required longer training and was less flexible compared to XGBoost.

Real-world Application

By enhancing the ability to detect fraudulent transactions, this approach helps reduce the risk of undetected fraud, thereby improving the overall security and reliability of financial transactions.

Thank You!